

ex. 1.1

The electrostatic force between two protons separated by 4×10^{-15} m is:

$$F = k_e \frac{q_1 q_2}{r^2} = 9 \times 10^9 \times \frac{1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{(4 \times 10^{-15})^2} = 14.4 \text{ N}$$

The gravitational force between two protons separated by 4×10^{-15} m is:

$$F = G \frac{m_1 m_2}{r^2} = 6.67 \times 10^{-11} \times \frac{1.67 \times 10^{-27} \times 1.67 \times 10^{-27}}{(4 \times 10^{-15})^2} = 1.2 \times 10^{-25} \text{ N}$$

ex. 2

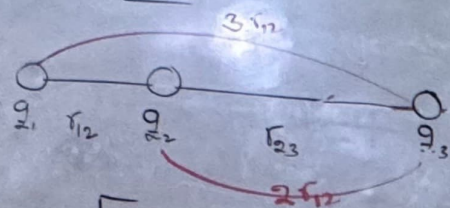
The charge should be placed on Earth and Moon to neutralize the gravitational force is :

$$\therefore \text{To neutralize} \rightarrow F_E = F_G$$

$$k_e \frac{q^2}{r^2} = G \frac{m_1 m_2}{r^2}$$

$$q^2 = \frac{G m_1 m_2}{k_e} = \frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times 7.36 \times 10^{22}}{9 \times 10^9} = 3.261 \times 10^{27} \rightarrow q = 5.711 \times 10^{13} \text{ C}$$

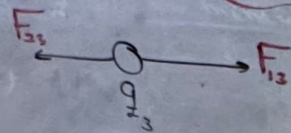
ex. 3



$$\frac{q_1}{q_2} = - \frac{r_{12}^2}{r_{23}^2} = - \frac{(3r_{12})^2}{(2r_{12})^2} = -2.25$$

From the figure shown, if $r_{23} = 2r_{12}$. The ratio q_1/q_2 required to make $F_t = 0$ on q_2 is:

$F_t = 0$ on q_2 :



$$F_{21} = -F_{23}$$

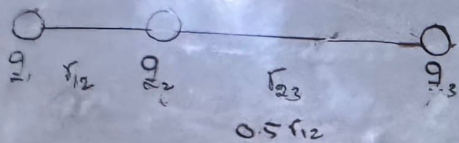
$$k \frac{q_1 q_2}{r_{12}^2} = -k \frac{q_2 q_3}{r_{23}^2}$$

$$\frac{q_1}{q_2} = - \frac{r_{12}^2}{r_{23}^2}$$

$$\rightarrow \frac{q_1}{q_2} = - \frac{r_{12}^2}{r_{23}^2} = - \frac{(3r_{12})^2}{(2r_{12})^2} = - \frac{9}{4} = -2.25$$

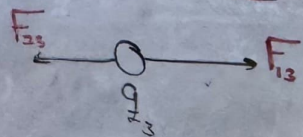
ex. 3

q₁



From the figure shown, if $r_{23} = r_{12}/2$. The ratio q_1/q_2 required to make $F_t = 0$ on q_3 is:

$F_t = 0$ on q_3 :



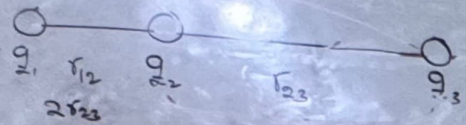
$$F_{13} = -F_{23}$$

$$k \frac{q_1 q_3}{r_{13}^2} = -k \frac{q_2 q_3}{r_{23}^2}$$

$$\frac{q_1}{q_2} = - \frac{r_{13}^2}{r_{23}^2}$$

$$\rightarrow \frac{q_1}{q_2} = - \frac{r_{12}^2}{r_{23}^2} = - \frac{(1.5 r_{12})^2}{(0.5 r_{12})^2} =$$

ex. 3



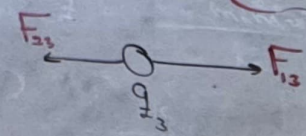
$$\underline{q_1}$$

$$\frac{(3r_{23})^2}{(r_{23})^2} = 9$$

From the figure shown, if $r_{12} = 2r_{23}$

The ratio q_1/q_2 required to make $F_{\text{net}} = 0$ on q_3 is:

$$F_{\text{net}} = 0 \text{ on } q_3:$$



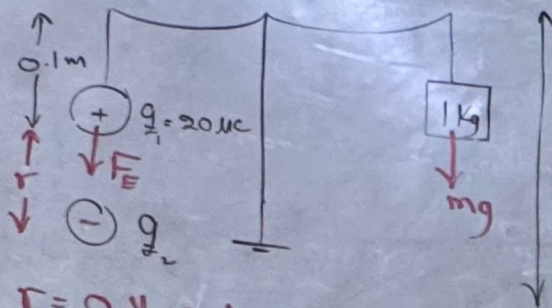
$$F_{13} = -F_{23}$$

$$k \frac{q_1 q_3}{r_{13}^2} = -k \frac{q_2 q_3}{r_{23}^2}$$

$$\frac{q_1}{q_2} = - \frac{r_{13}^2}{r_{23}^2}$$

$$\rightarrow \frac{q_1}{q_2} = - \frac{r_{12}^2}{r_{23}^2} = - \frac{(1.5r_{23})^2}{(0.5r_{23})^2} = -9$$

ex. 4



$$r = 0.4 - 0.1 = 0.3 \text{ m}$$

The value of q to balance is:

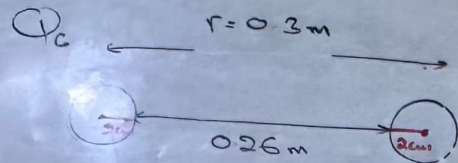
∴ To balance

$$F_E = mg$$

$$\frac{k_e q_1 q_2}{r^2} = mg$$

$$q_2 = \frac{mg \times r^2}{k_e q_1} = \frac{1 \times 10 \times 0.3^2}{9 \times 10^9 \times 20 \times 10^{-6}} = 5 \times 10^{-6} \text{ C}$$

$$q_2 = -5 \mu\text{C}$$



Two spheres of 2 cm radius, the repulsive force is 1500 mN, if the total charges are 8 μC .
The charge on each sphere is.

$$r = 0.26 + 2 \times 10^{-2} + 2 \times 10^{-2} = 0.3 \text{ m}$$

$$q_1 + q_2 = 8 \mu\text{C}$$

$$q_2 = 8 \mu\text{C} - q_1$$

$$F_e = k_e \frac{q_1 q_2}{r^2}$$

$$F_e = k_e \frac{q_1 q_2}{r^2} \rightarrow 1500 \times 10^{-3} = \frac{9 \times 10^9}{0.3^2} q_1 (8 \times 10^{-6} - q_1)$$

$$\frac{1500 \times 10^{-3} \times 0.3^2}{9 \times 10^9} = 8 \times 10^{-6} q_1 - q_1^2$$

$$q_1^2 - 8 \times 10^{-6} q_1 + 1.5 \times 10^{-11} = 0$$

$q_1 = 5 \mu\text{C}$

$$q_1 = 5 \mu\text{C} \rightarrow q_2 = 3 \mu\text{C}$$

$$\frac{1500 \times 10^{-3} \times 0.3^2}{9 \times 10^9} = 8 \times 10^{-6} q_1 - q_1^2 \rightarrow 1.5 \times 10^{-11} = 8 \times 10^{-6} q_1 - q_1^2$$